

Alpha test plan

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1. Target performance metrics

The goal of testing the alpha version of the device will be to determine if the target values for key specifications can be met with our circuit designs.

Performance metric	Target value	Explanation
Output frequency	25 MHz	Measured with an RF spectrum analyzer.
Output power	≥ 30 dBm differential	Measured with an RF spectrum analyzer. 30 dBm differential = 27 dBm per electrode.
Static current	< 10 mA	Current consumed when the RF signal is not being transmitted. Needed for 12 hr battery life with a 140 mAh battery.
Tx current	< 500 mA	Current consumed when the RF signal is being transmitted.
EMG noise floor	< 0.5 mV	EMG spikes reach up to +5 mV, which would give us an SNR of 10 dB.
EMG sampling rate	≥ 1000 samples/s	Enough to measure signals up to 500 Hz.

2. MCU circuit validation

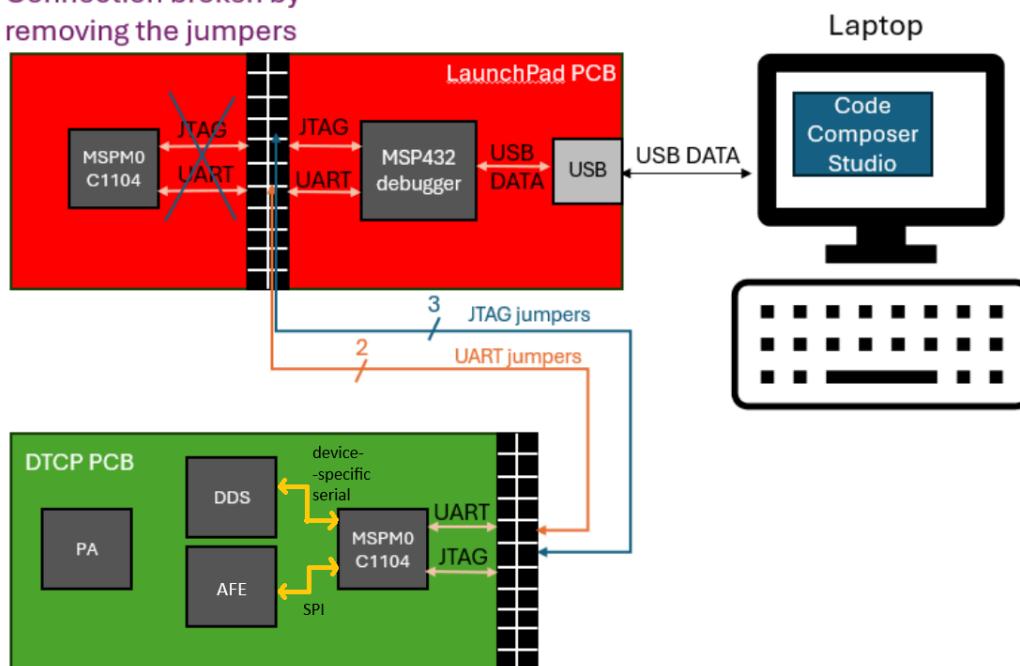
a. Expected behavior

After assembling the microcontroller circuit and making the connections to the programmer/debugger, the device is recognized by the CCS IDE, and it can be flashed with firmware.

b. Test procedure

Set up the devices as follows:

Connection broken by removing the jumpers



Connect the TMS, TCK, and RST lines from the Launchpad to the corresponding pins on the PCB. Connect 3.3V power from a DC power supply to the corresponding power port on the PCB. Open CCS on the laptop connected to the Launchpad, then click “Debug project” on any firmware project and observe if the debugging launches correctly and the code execution can be paused and continued.

3. RF circuit validation

a. Expected behavior

For the class-A PA circuit:

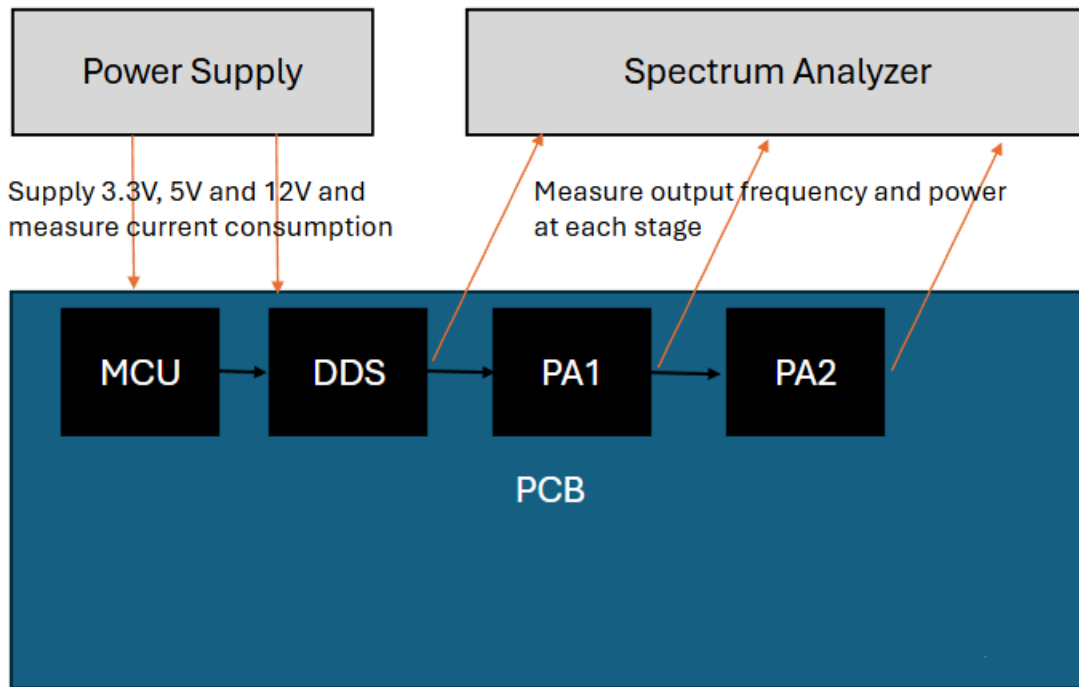
- DDS output: 0 ~ 10 dBm @ 25 MHz
- PA1 output: 5 ~ 15 dBm @ 25 MHz
- PA2 output: 30 ~ 33 dBm @ 25 MHz

For the class-E PA circuit

- 3.3 Vpp 25 MHz square wave at the DDS output
- 5 Vpp 25 MHz square wave at the PA1 (Gate Driver) output
- 30 - 35 dBm @ 25 MHz at the PA2 output

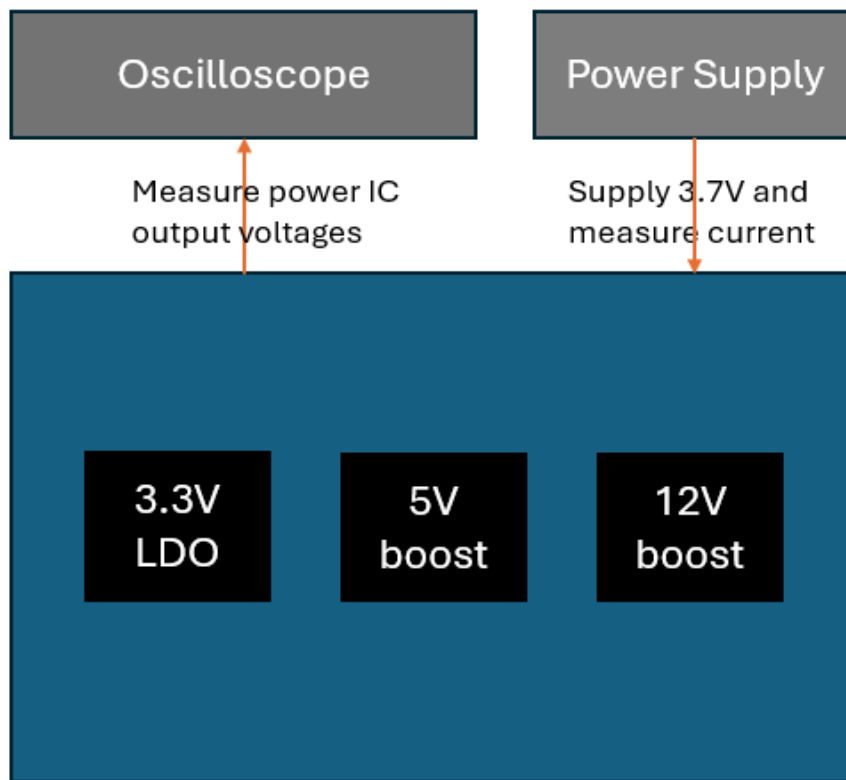
b. Test procedure

Set up equipment as follows:



Measure outputs in the order of the signal chain: DDS, then PA1, then PA2. Test each stage by powering it with a DC power supply, taking note of the initial static current. Then activate the DDS output and take note of the PA output power and frequency.

4. Power circuit validation
 - a. Expected behavior
The LDO and two buck converters output within 0.05V of the target output voltage.
 - b. Test procedure
Set up the equipment as follows:



Use a DC power supply in place of the 3.7V battery, and measure the LDO and boost converters' output voltages with an oscilloscope.

5. EMG recording validation

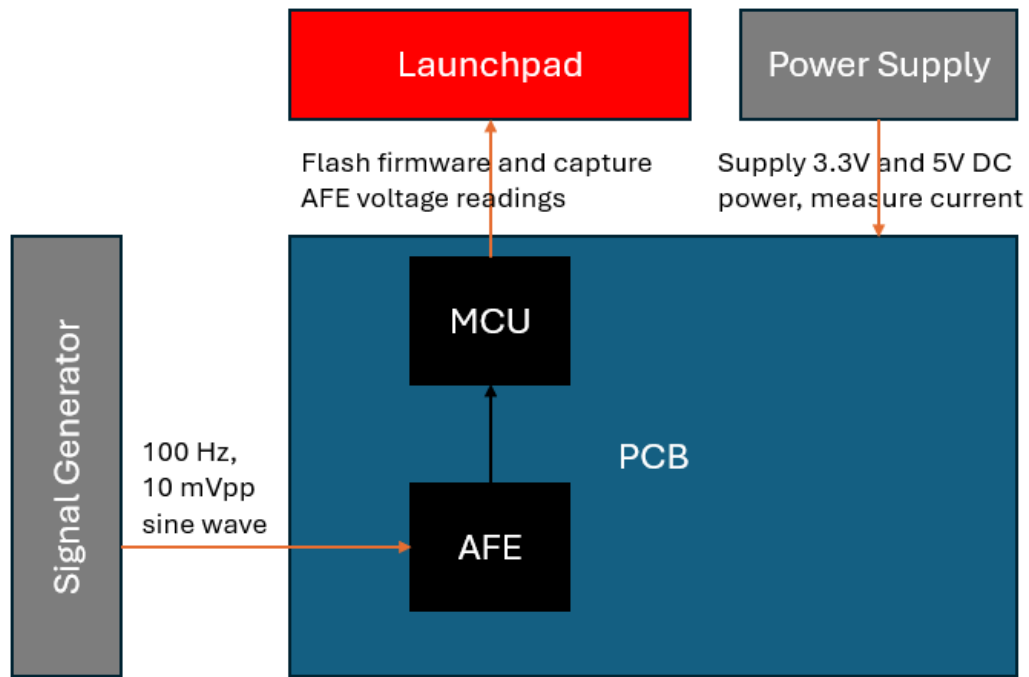
a. Expected behavior

The AFE responds to SPI commands, returning correct values in accordance with the datasheet. The SPI waveforms investigated with an oscilloscope are in line with the timing diagrams from the datasheet.

The measured voltage values recreate the waveform and the amplitude of the channel input and signal-to-noise (SNR) ratio is 10 dB or higher, meaning that the noise floor is below 0.5 mV.

Data ready signal is asserted at the rate of ~ 1 kHz (equivalent to the sampling rate)

b. Test procedure



Use power supply to supply 3.3V and 5V DC for the digital and analog parts of the board. Use a function generator to supply a 100 Hz, 10 mVpp sine wave to the input of a channel. Debug the firmware project using CCS and use the Graph functionality to continuously plot the recorded values. Short the inputs of the channel and record the channel data to measure the noise. Use the oscilloscope to measure the SPI signals and the frequency of the falling edge of the data ready signal.